250919-Application of Glasso on INT-Transformed Zero-Inflated Data

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0. Last Time

1. Pearson partial correlation is more sensitive to transformation than Spearman and Kendall
2. Normality satisfied after INT transformation.
3. We can experiment Glasso on INT data, and compare the result with network given by Spearman and Kendall
4. Multiple test correction for p-value based edge selection should be tested.

1. Today

1. Application of Glasso on zero-inflated data before and after INT transformation

- Compare the networks before and after transformations

2. Loading Data and Packages

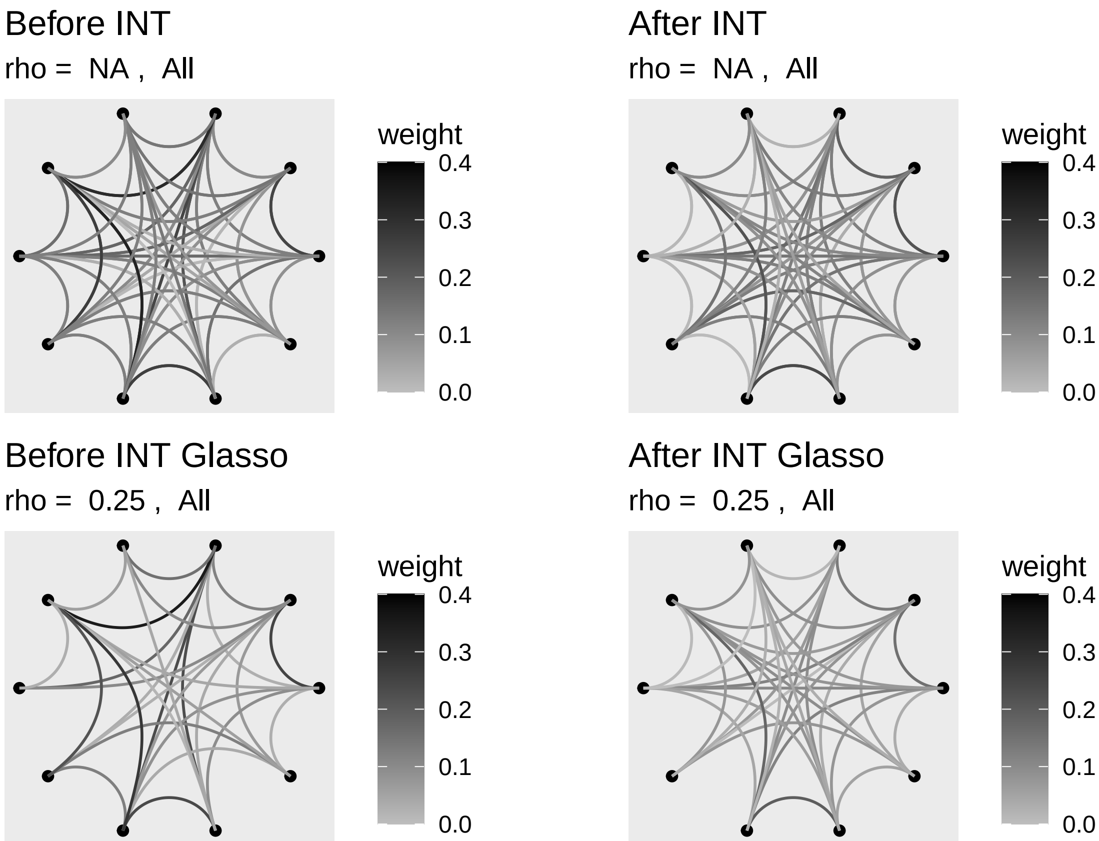
```
## Load packages
library(tidyverse)
library(ggplot2)
library(cowplot)
library(igraph)
library(tidygraph)
library(ggraph)
library(ppcor)

## Load customized functions
source("../R-scripts/INT.R")

## Load and Sample Methylation Data
X.T <-  readRDS(file = "../../Data/Common_pan_cancer_hyper_bins_adjusted_and_normalized_cnt_in_SE.RDS")
set.seed(226)
sampDMR <- sample(rownames(X.T), 10) ## Sample only 10 DMRs for speed sake
original.X <- t(X.T[sampDMR, ])

## Load Cancer Types
sampleInfo <- read.csv("../../Data/Common_pan_cancer_hyper_bins_adjusted_cnt_in_SE_samples_info.csv")
cancerTypes <- sampleInfo$cancer_type
```

```r
## INT Transformation
INT.X <- INT(original.X, 0.25)
```

3. Glasso on Zero-Inflated the Data, Before and After INT, All Samples

```r
source("../R-scripts/graphMaker.R")
library(glasso)
## Partial Pearson Correlation before any transformations
ParCorrNone <- pcor(original.X, method = "pearson")

## Partial Pearson Correlation after INT transformations
ParCorrINT <- pcor(INT.X, method = "pearson")

## Glasso before INT
glassoNone <- glassoPreMat(original.X, rho = 0.25)
ParCorrGlassoNone <- parCorr_from_precision(glassoNone$wi, 0.25, nrow(original.X))

## Glasso after IVT
glassoINT <- glassoPreMat(INT.X, rho = 0.25)
ParCorrGlassoINT <- parCorr_from_precision(glassoINT$wi, 0.25, nrow(INT.X))

## Graphical Comparison
ParCorrNoneGraph <- plot_parcorr_graph(ParCorrNone$estimate, NA, "All", "Before INT")
ParCorrINTGraph <- plot_parcorr_graph(ParCorrINT$estimate, NA, "All", "After INT")
ParCorrNoneGlassoGraph <- plot_parcorr_graph(ParCorrGlassoNone, 0.25, "All", "Before INT Glasso")
ParCorrINTGlassoGraph <- plot_parcorr_graph(ParCorrGlassoINT, 0.25, "All", "After INT Glasso")

plot_grid(plotlist = list(ParCorrNoneGraph, ParCorrINTGraph,
                          ParCorrNoneGlassoGraph, ParCorrINTGlassoGraph),
          nrow = 2, ncol = 2)
```

```
## Ignoring unknown labels:
## * colour : "Pcorr"
## Ignoring unknown labels:
## * colour : "Pcorr"
## Ignoring unknown labels:
## * colour : "Pcorr"
## Ignoring unknown labels:
## * colour : "Pcorr"
```




```
ggsave('../../Images/GraphVis/AllSampleINTGlasso.png')
```

```
## Saving 6.5 x 4.5 in image
```

Main Takeaway: INT transformation can lead to slightly differences in partial correlations.

Observations: 1. For unpenalized Pcorr graphs, egde sets look similar before and after INT, but the color lightens after INT. 2. For glasso graphs, not only color lightens, but also removal of heavily weighted old edges and replacement by new and lighter weighted edges.

4. Glasso on Zero-Inflated the Data, Before and After INT, Cancer Samples

```
## Partial Pearson Correlation before any transformations
ParCorrNone <- pcor(original.X[str_ends(cancerTypes, "Cancer"),], method = "pearson")

## Partial Pearson Correlation after INT transformations
ParCorrINT <- pcor(INT.X[str_ends(cancerTypes, "Cancer"),], method = "pearson")

## Glasso before INT
glassoNone <- glassoPreMat(original.X[str_ends(cancerTypes, "Cancer"),], rho = 0.25)
ParCorrGlassoNone <- parCorr_from_precision(glassoNone$wi, 0.25, nrow(original.X))

## Glasso after IVT
```

```r
glassoINT <- glassoPreMat(INT.X[str_ends(cancerTypes, "Cancer"),], rho = 0.25)
ParCorrGlassoINT <- parCorr_from_precision(glassoINT$wi, 0.25, nrow(INT.X))

## Graphical Comparison
ParCorrNoneGraph <- plot_parcorr_graph(ParCorrNone$estimate, NA, "Any Cancer", "Before INT")
ParCorrINTGraph <- plot_parcorr_graph(ParCorrINT$estimate, NA, "Any Cancer", "After INT")
ParCorrNoneGlassoGraph <- plot_parcorr_graph(ParCorrGlassoNone, 0.25, "Any Cancer", "Before INT Glasso")
ParCorrINTGlassoGraph <- plot_parcorr_graph(ParCorrGlassoINT, 0.25, "Any Cancer", "After INT Glasso")

plot_grid(plotlist = list(ParCorrNoneGraph, ParCorrINTGraph,
                          ParCorrNoneGlassoGraph, ParCorrINTGlassoGraph),
          nrow = 2, ncol = 2)
```

```
## Ignoring unknown labels:
## * colour : "Pcorr"
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## * colour : "Pcorr"
## Ignoring unknown labels:
## * colour : "Pcorr"
```

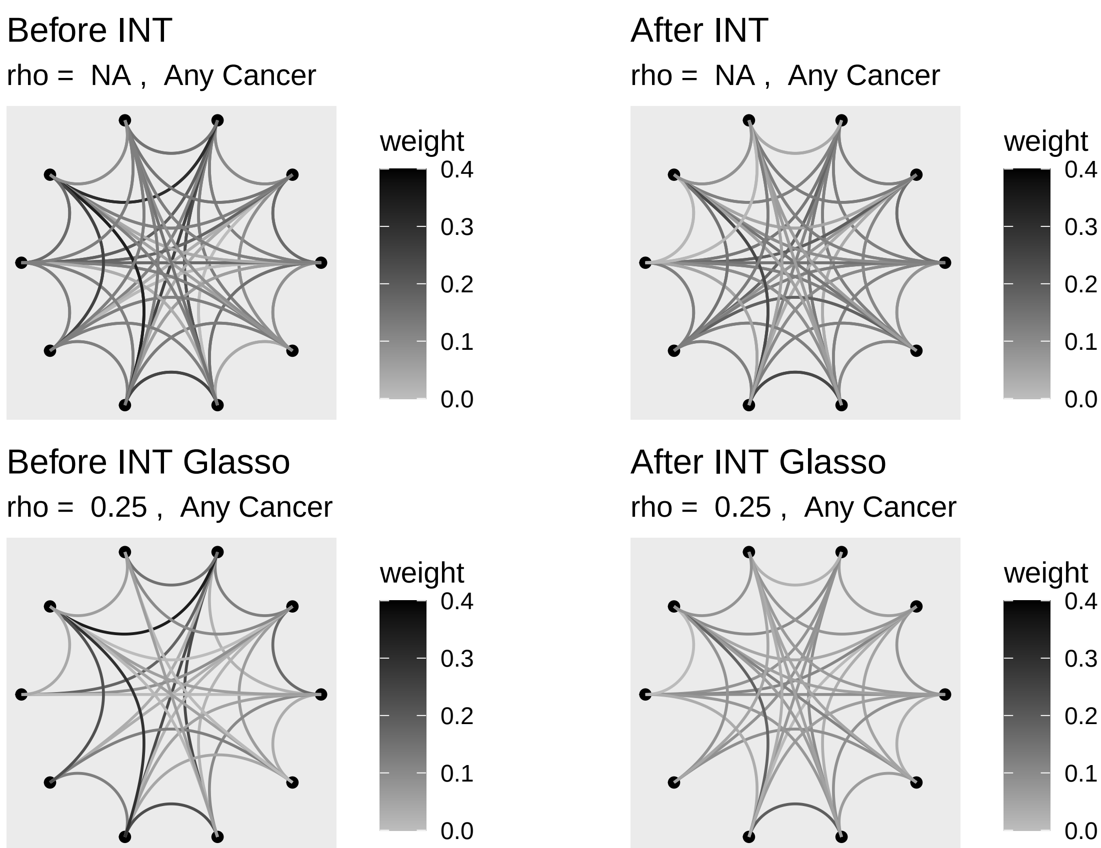


```r
ggsave('../../Images/GraphVis/CancerINTGlasso.png')
```

```
## Saving 6.5 x 4.5 in image
```

Main Takeaway: Partial correlations of methylation between gene regions are strong in cancer individuals, and dominate the overall methylation.

Observations: Graphs of cancer samples look similar to partial correlation graphs based on all samples, with a few edges create and removed relative to the reference graph.

5. Glasso on Zero-Inflated the Data, Before and After INT, Normal Samples

```
## Partial Pearson Correlation before any transformations
ParCorrNone <- pcor(original.X[cancerTypes == "Normal", ], method = "pearson")

## Partial Pearson Correlation after INT transformations
ParCorrINT <- pcor(INT.X[cancerTypes == "Normal", ], method = "pearson")

## Glasso before INT
glassoNone <- glassoPreMat(original.X[cancerTypes == "Normal", ], rho = 0.25)
ParCorrGlassoNone <- parCorr_from_precision(glassoNone$wi, 0.25, nrow(original.X))

## Glasso after IVT
glassoINT <- glassoPreMat(INT.X[cancerTypes == "Normal", ], rho = 0.25)
ParCorrGlassoINT <- parCorr_from_precision(glassoINT$wi, 0.25, nrow(INT.X))

## Graphical Comparison
ParCorrNoneGraph <- plot_parcorr_graph(ParCorrNone$estimate, NA, "Healthy Normal", "Before INT")
ParCorrINTGraph <- plot_parcorr_graph(ParCorrINT$estimate, NA, "Healthy Normal", "After INT")
ParCorrNoneGlassoGraph <- plot_parcorr_graph(ParCorrGlassoNone, 0.25, "Healthy Normal", "Before INT Gla
ParCorrINTGlassoGraph <- plot_parcorr_graph(ParCorrGlassoINT, 0.25, "Healthy Normal", "After INT Glasso

plot_grid(plotlist = list(ParCorrNoneGraph, ParCorrINTGraph,
                          ParCorrNoneGlassoGraph, ParCorrINTGlassoGraph),
          nrow = 2, ncol = 2)
```

```
## Ignoring unknown labels:
## * colour : "Pcorr"
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## Ignoring unknown labels:
## * colour : "Pcorr"
## Ignoring unknown labels:
## * colour : "Pcorr"
```

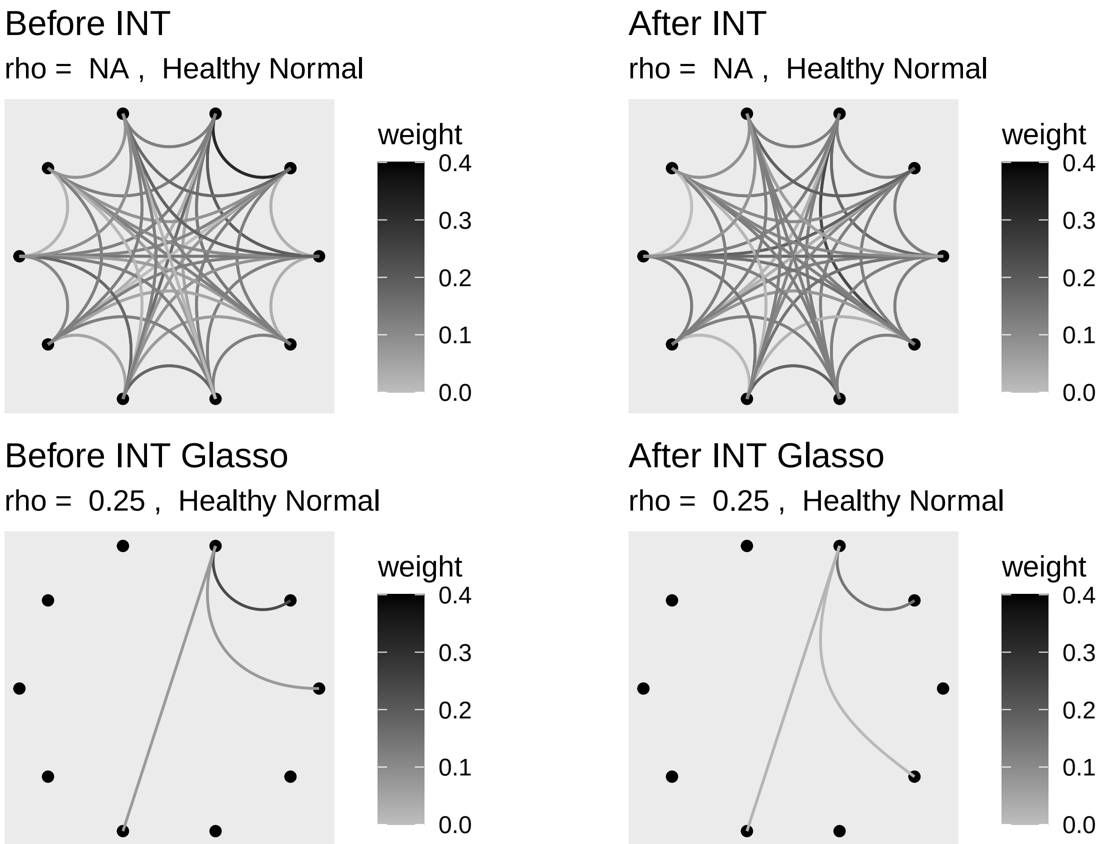


```r
ggsave('../../Images/GraphVis/HealthyINTGlasso.png')
```

```
## Saving 6.5 x 4.5 in image
```

Main Takeaways: Partial Correlations of healthy Normals are much weaker than cancer samples.

6. HurdleNormal on INT Transformed Data

```r
# library(HurdleNormal)
## Glasso before INT


## Glasso after IVT


## Graphical Comparison
```